

What is machine learning?

machine learning is the program that can improve performance from the experience.

How many types of ML?

- Supervisor learning: learning with the dataset that is labeled
- NonSupervisor learning: learning with the dataset that is NOT labeled
- Reinforcement learning: learn from environment, which means that they get score depending on their choices.

+ overfitting: the phenomenon that the model has the high accuracy with training set but not fit with the test set. It is because the model ~~passing~~ ~~the training~~ all training value in ~~it~~ leading to they do not fit with the testing value.

The reason is the model is too complicated

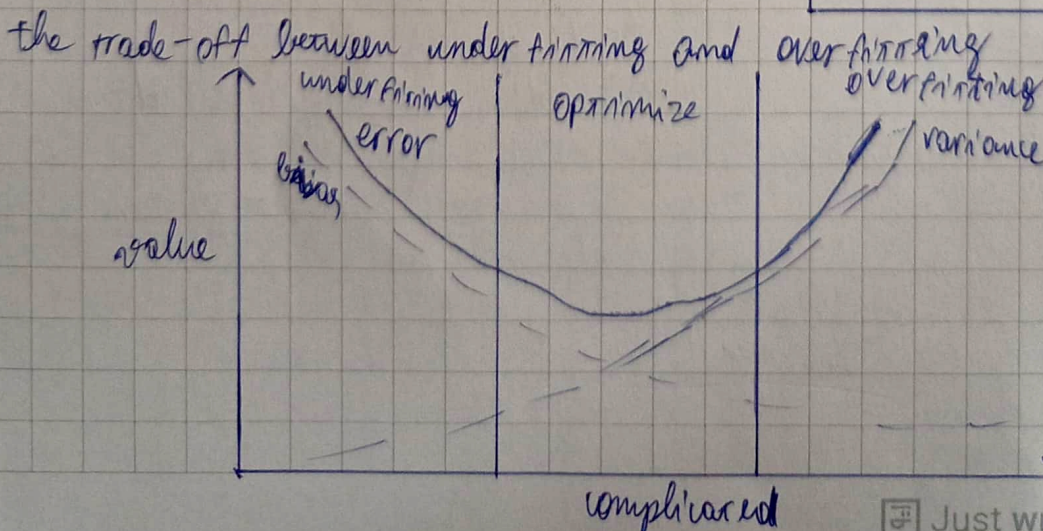
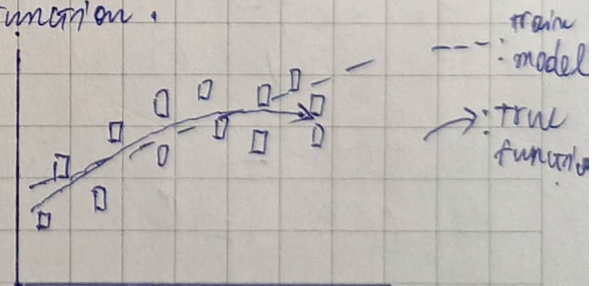
→ the model would be flexible to fit all training set.

⇒ low Bias but high Variance.



+ underfitting: is the phenomenon that the model cannot fit with the true ~~function~~ ^{function}. It is because the model is simpler than the true function. For example, the true function is a curve but the model is linear, leading to the model never fitting with the true function.

⇒ high Bias ~~but~~ ^{low} variance



Entropy is the value that describe the chaos of the dataset.

$$H(S) = - \sum_{i=1}^k p_i \log_2(p_i)$$

Information gain is the value that symbolic the reduction of entropy in the dataset of an attribute which be chosen for the node of decision tree.

$$IG(S, A) = H(S) - \sum_{v \in A} \frac{|S_v|}{|S|} H(S_v)$$

How do we choose the attribute to become a root or internal node?
 → we ~~need~~ choose the attribute that can help the entropy of dataset reduce as much as possible. It is because reduce ^{more} ~~much~~ entropy mean the data become ~~more~~ purer. Particularly, the attribute have the highest information gain mean that this attribute affect ~~more~~ seriously to the result. ⇒ we ~~the~~ choose the attribute that have the highest IG .

Gini index: $Gini(S) = 1 - \sum_{i=1}^k p_i^2$

Gini index have the min value = 0 (pure) and max value = 0.5 (not pure). We choose the attribute that have the smaller Gini index to become the root or internal root. It is because if we choose the attribute that have the higher value of Gini index leading to the separate of the label will split uniformly.

In the continuous attribute, the first we ~~the~~ sort that attribute, then we use the mean of each 2 adjacent value as the ~~node~~ threshold, then calculate it by using Gini or entropy and information gain formula. Which threshold have the best value will be chosen.

There are 2 way to prevent decision tree get overfitting.

1. ~~post~~ - pruning: we prune tree ~~and~~ while plan the tree, we can set some rule for the tree like: max-depth; ~~the~~ number of sample minimum sample can be chosen, ...; and we Gain score to ~~measure~~ evaluate the quality of decision tree.
2. pre-pruning: we build the ~~whole~~ tree then pruning,